Network Diagnostic Tools Assignment Report

1. Ping Test to www.youtube.com

Sample result: Average response time = 24 ms. Replace with your actual ping result from your computer.

2. Traceroute to www.flipkart.com

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Hop 1: Home Router (192.168.1.1)
Hop 2: ISP Gateway
Hop 3: ISP Core Router
Hop 4: Regional Backbone Router
Hop 5: Destination Network
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Replace the above hops with your actual tracert/traceroute output.

3. NSLookup for www.zomato.com

Record the IP address returned by nslookup. Insert your actual lookup result here.

4. Local Network Ping Test

When the target device is online, ping receives replies. If the device is offline, requests typically time out or return destination host unreachable.

5. Summary Report

Ping is useful for checking whether a device or website is reachable and measuring response time. Traceroute is useful for diagnosing slow connections by identifying delays along the network path. NSLookup is helpful when DNS problems are suspected because it verifies domain-name-to-IP resolution. Example: use ping for YouTube connectivity, traceroute for Flipkart route analysis, and nslookup for Zomato DNS verification.